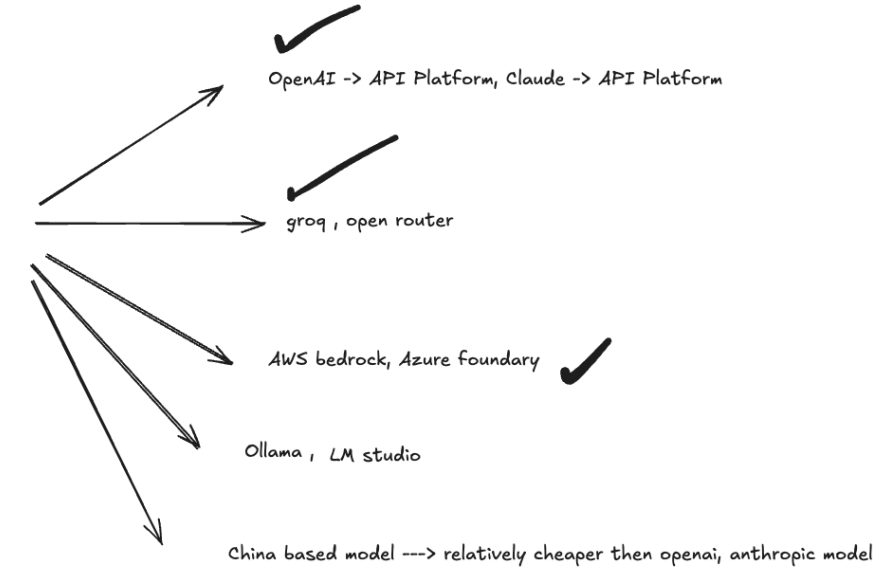
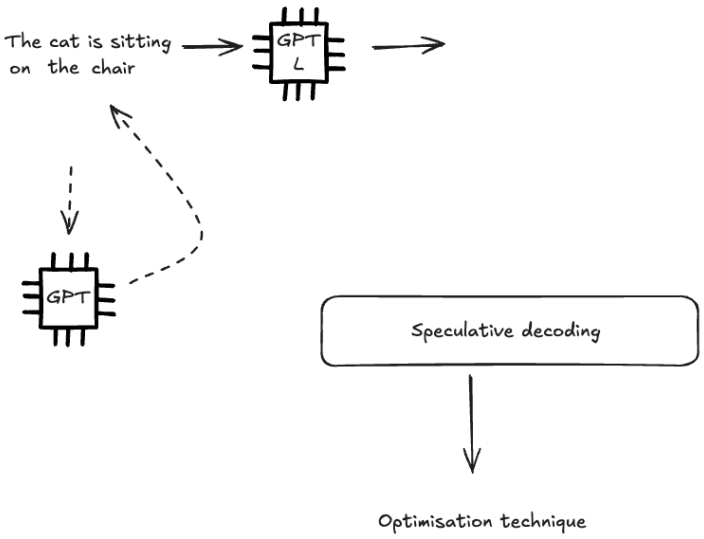
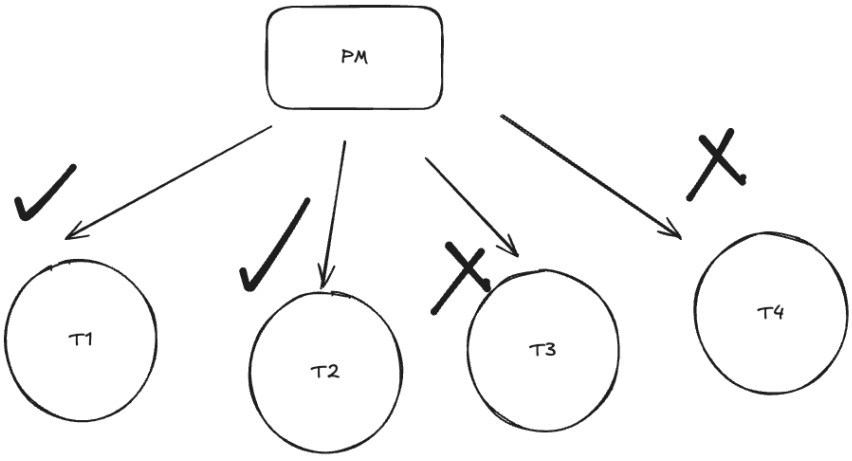
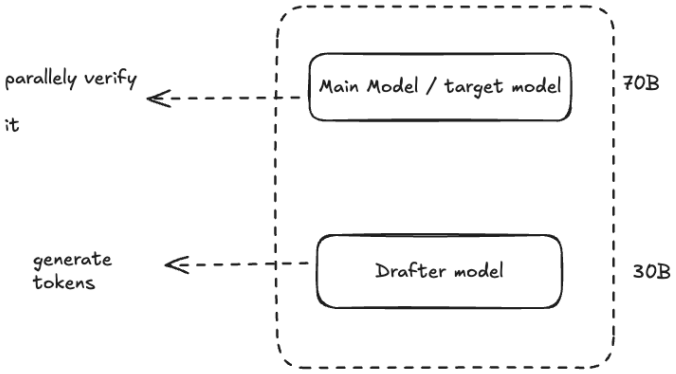

OpenAI -> API Platform, Claude -> API Platform
groq , open router
AWS bedrock, Azure foundary
Ollama , LM studio
China based model ---> relatively cheaper then openai, anthropic model


The cat is sitting on the chair
GPT
L
GPT
Speculative decoding
Optimisation technique


PM
T1
T2
T3
T4



Speculative decoding is an inference optimisation technique where a smaller and a faster model predicts a set of future tokens, while the larger target model verifies those prediction efficiently.

Normal autoregressive model

Prompt -> LLM Forward pass -> generate token 1 -> LLM forward pass ->
Generate token 2 -> LLM forward pass -> Token 3

Forward pass -> means running the input tokens through the neural network

probability distribution

Decoding Phase -> we determine which token to finally choose

The cat is

sleeping x
sitting y
playing z
speaking ...
....

greedy decoding -> pick the one with highest probability

70B parameter ->

T1 -> LLM -> T2

T1 + T2 -> LLM. -> T3

T1 + T2 + T3 -> LLM. -> T4

10B parameter ->

T1 -> LLM -> T2

T1 + T2 -> LLM. -> T3

T1 + T2 + T3 -> LLM. -> T4

parallel computation
to the target model

time per generated token
cost per generated token

1. Draft token generation

2. Parallel verification

3. Rejection

target probability < draft probability -> reject the oken

Target probability >= draft probability -> accept

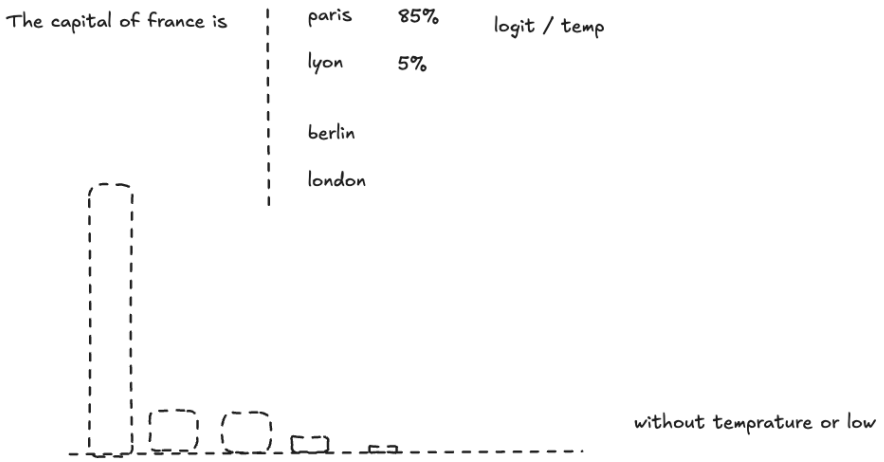
-> Sampling

-> Instead of taking highest probability, we try to sample according the probability distribution.

Using tempreature

doesnt change what model has already learned.

it impacts how the next token during the inference is selected.



1. Prompt engineering

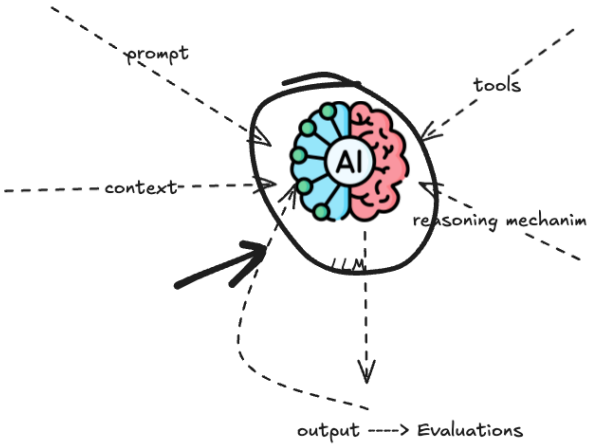
2. Context engineering

RAG

internally company audit done by llm

documents, db details and more ....

Harness engineering



AGENTS

Claude Code --> Coding agent running on top of Claude models like opus

- u have capability of dividing the task into smaller subtask
- fetch the relevant information from the codebase
- we can spin sub agents
- evaluating output of the subagents
- verifiy that code by running existing tests

security

Memory

. . . .

opus + ... lot of more things